

LAB 02

Title: Checking the Validity of an Input String Using C++ with Debugging Techniques

Objective

To learn how to debug a C++ program using Breakpoints and the Watch Window, and to determine whether an input string is valid based on predefined conditions.

Procedure

1. Complete environment setup.
2. Write the input string validation code.
3. Set break points.

Breakpoint Location	Reason
getline(cin, str);	To verify that the complete input string is read correctly.
for(int i = 0; i < str.length(); i++)	To observe each character of the input string.
if(isalpha(str[i]))......	To check the condition execution.
else	To identify the first invalid character and terminate the validation process.

4. Execute with Next Line.
5. Track the changes on Watch.

Code with break-points

```
StringValidity.cpp
1  #include <iostream>
2  #include <cctype>
3  using namespace std;
4
5  int main()
6  {
7      string str;
8
9      cout << "Enter a string: ";
10     getline(cin, str);
11
12     bool valid = true;
13
14     for(int i = 0; i < str.length(); i++)
15     {
16         if(isalpha(str[i]) || isdigit(str[i]) || str[i] == ' ')
17         {
18             continue;
19         }
20         else
21         {
22             valid = false;
23             break;
24         }
25     }
```

```

26
27     if(valid)
28     {
29         cout << "Valid String";
30     }
31     else
32     {
33         cout << "Invalid String";
34     }
35
36     return 0;
37 }

```

Output

